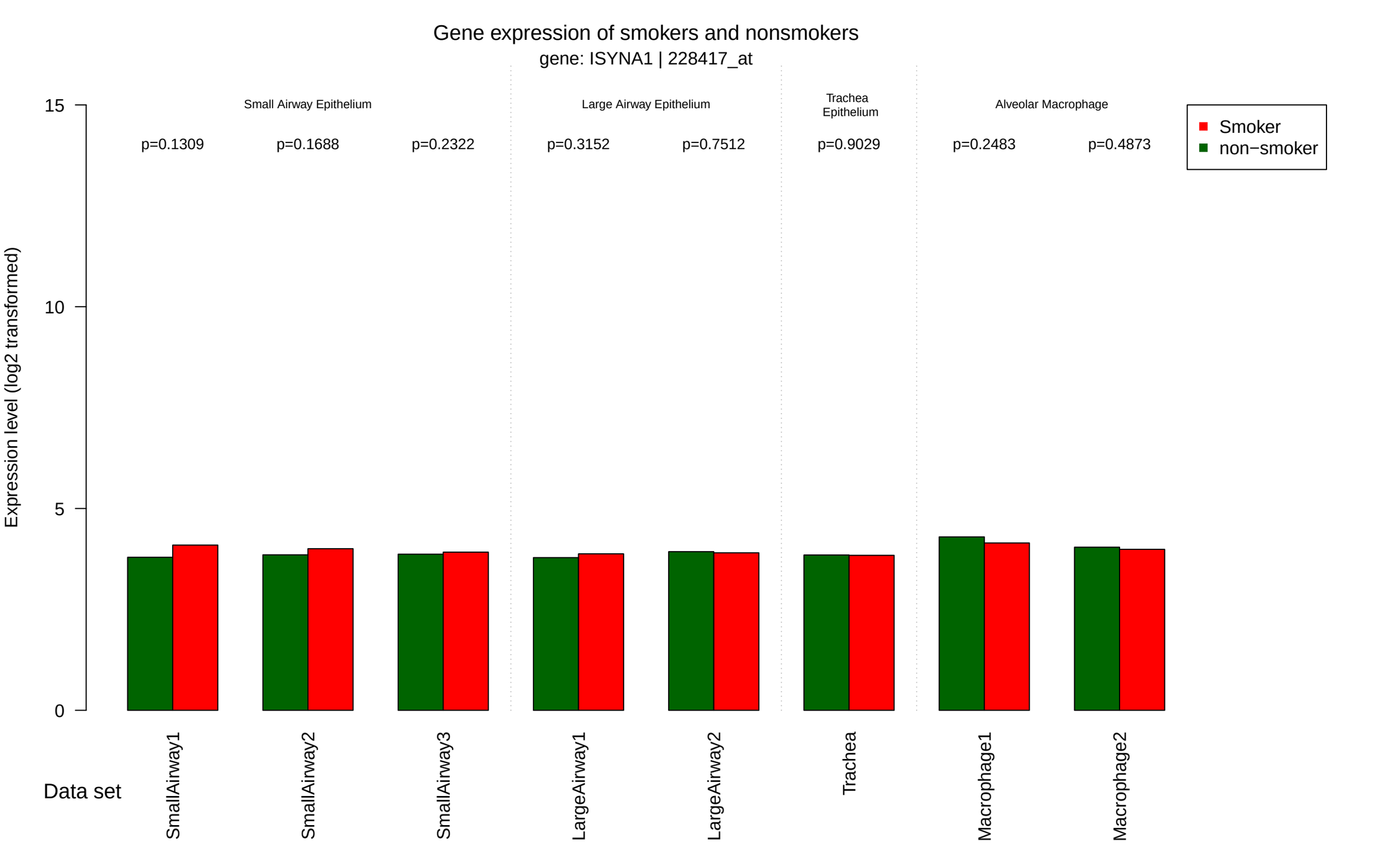
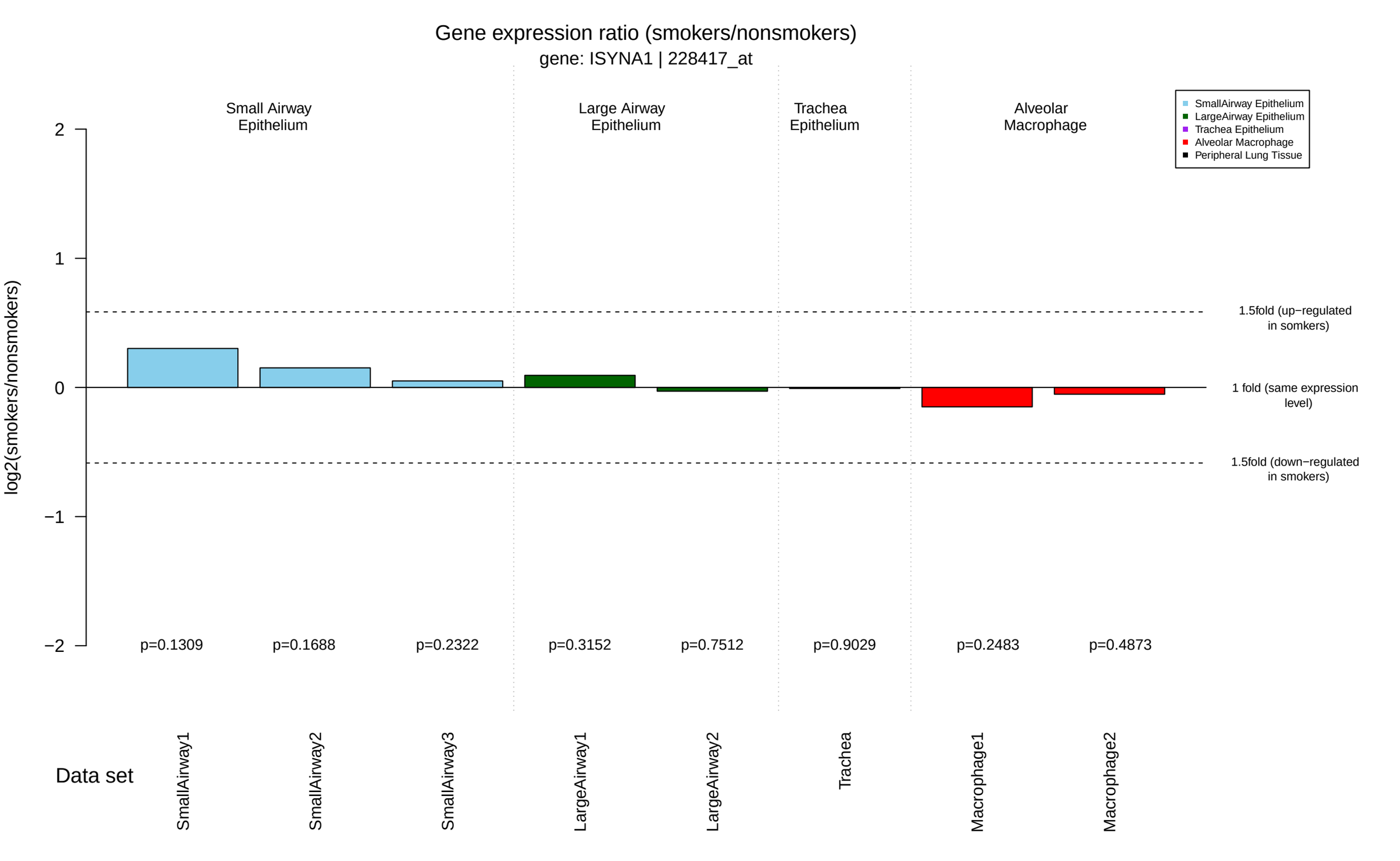
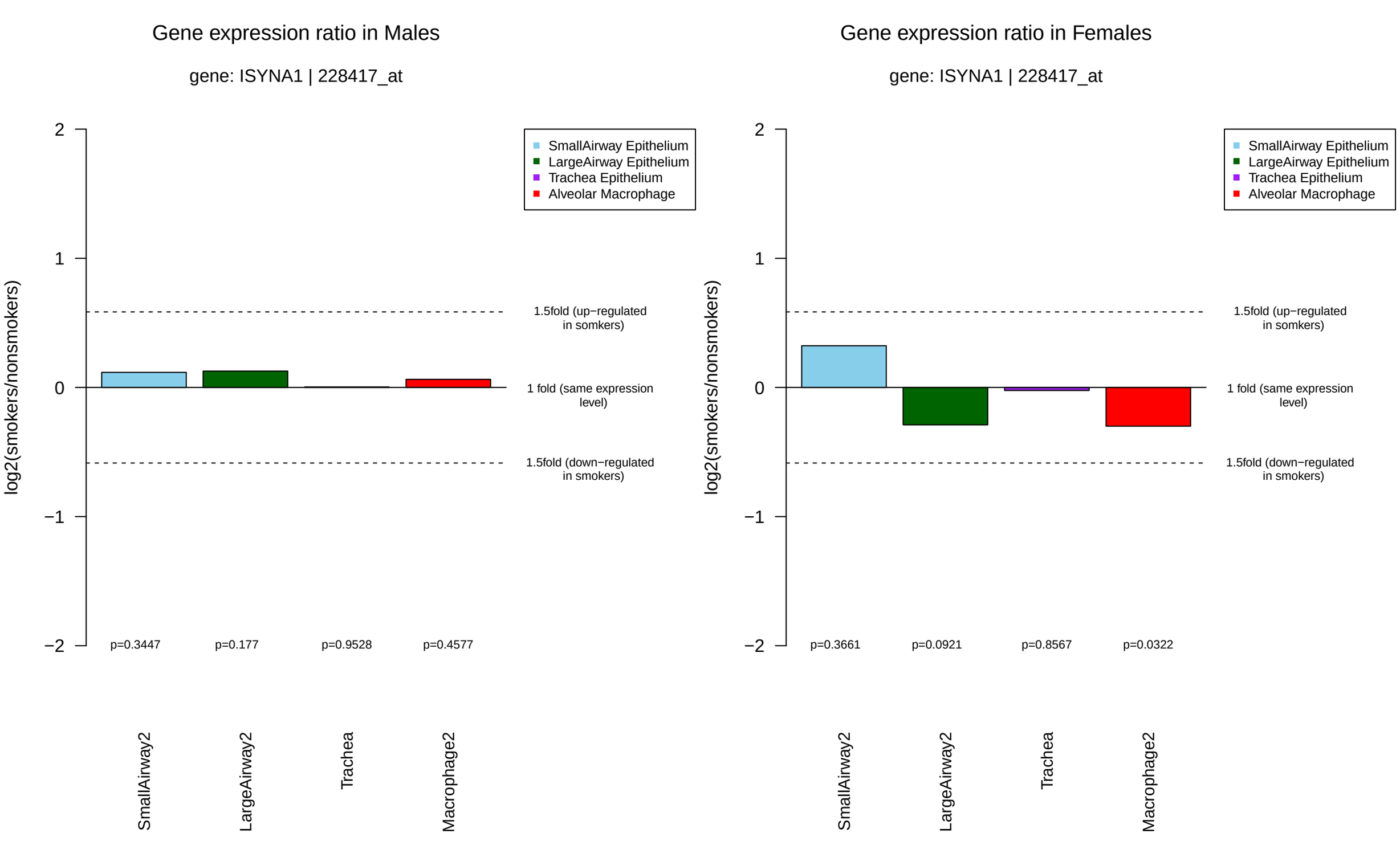
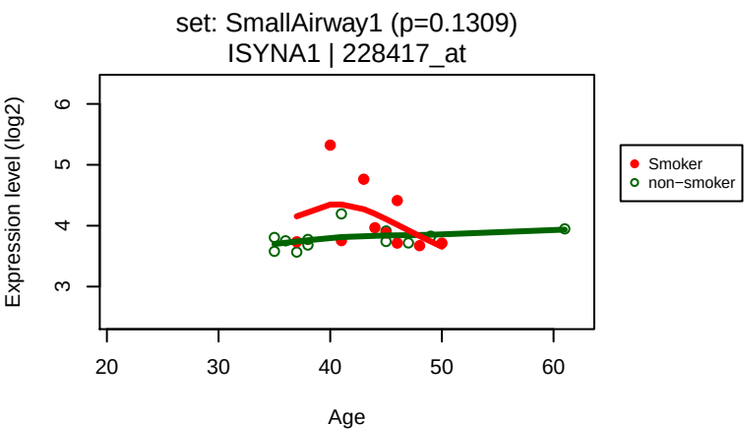
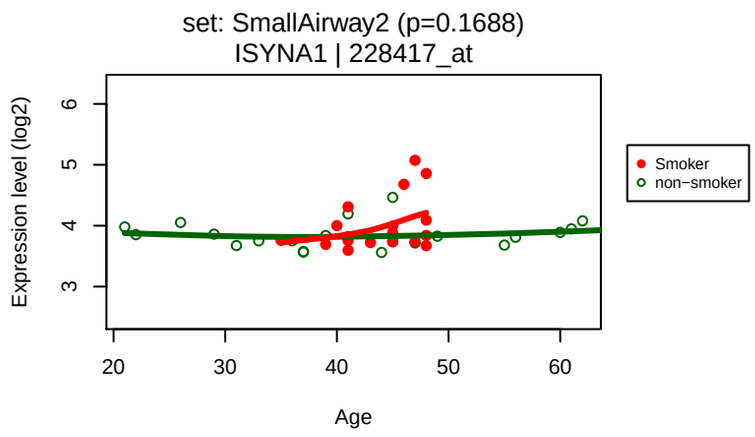
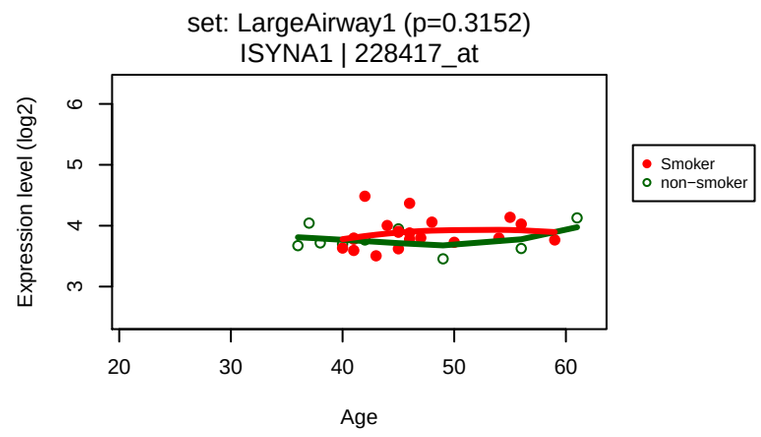

Gene expression of smokers and nonsmokers
gene: ISYNA1 | 228417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1309
p=0.1688
p=0.2322
p=0.3152
p=0.7512
p=0.9029
p=0.2483
p=0.4873
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

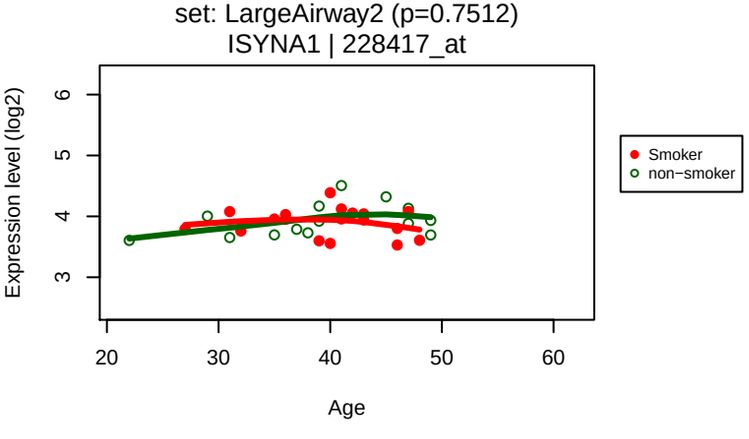

Gene expression ratio (smokers/nonsmokers)
gene: ISYNA1 | 228417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1309
p=0.1688
p=0.2322
p=0.3152
p=0.7512
p=0.9029
p=0.2483
p=0.4873
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

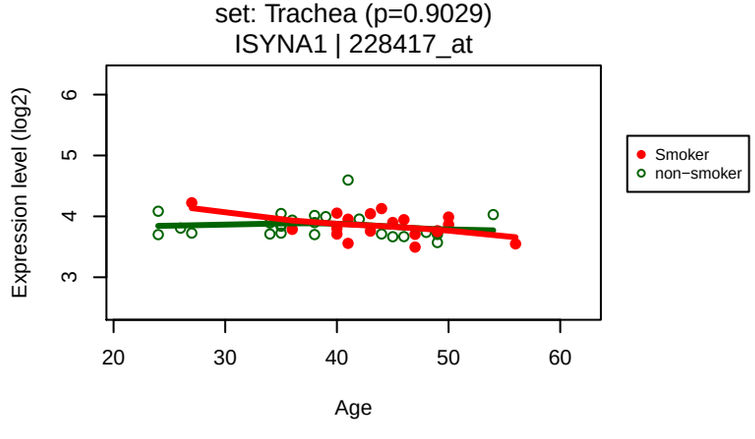

Gene expression ratio in Males
gene: ISYNA1 | 228417_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3447
p=0.177
p=0.9528
p=0.4577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISYNA1 | 228417_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3661
p=0.0921
p=0.8567
p=0.0322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

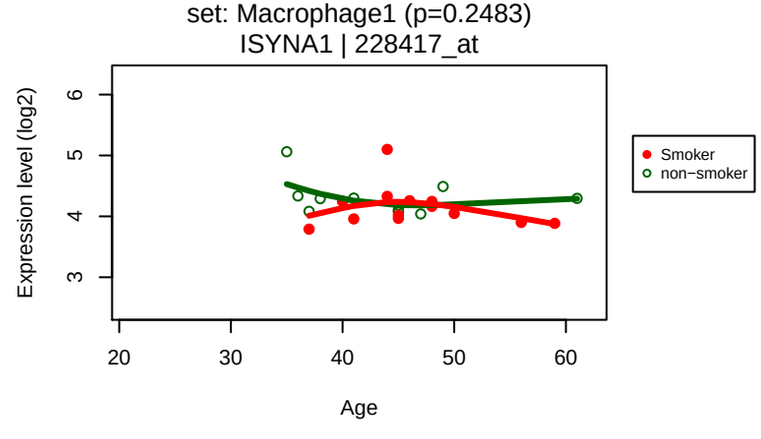

set: SmallAirway1 (p=0.1309)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker

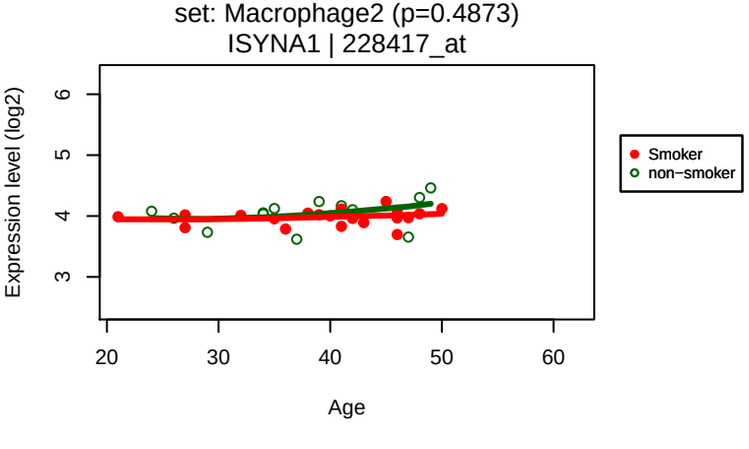

set: SmallAirway2 (p=0.1688)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker

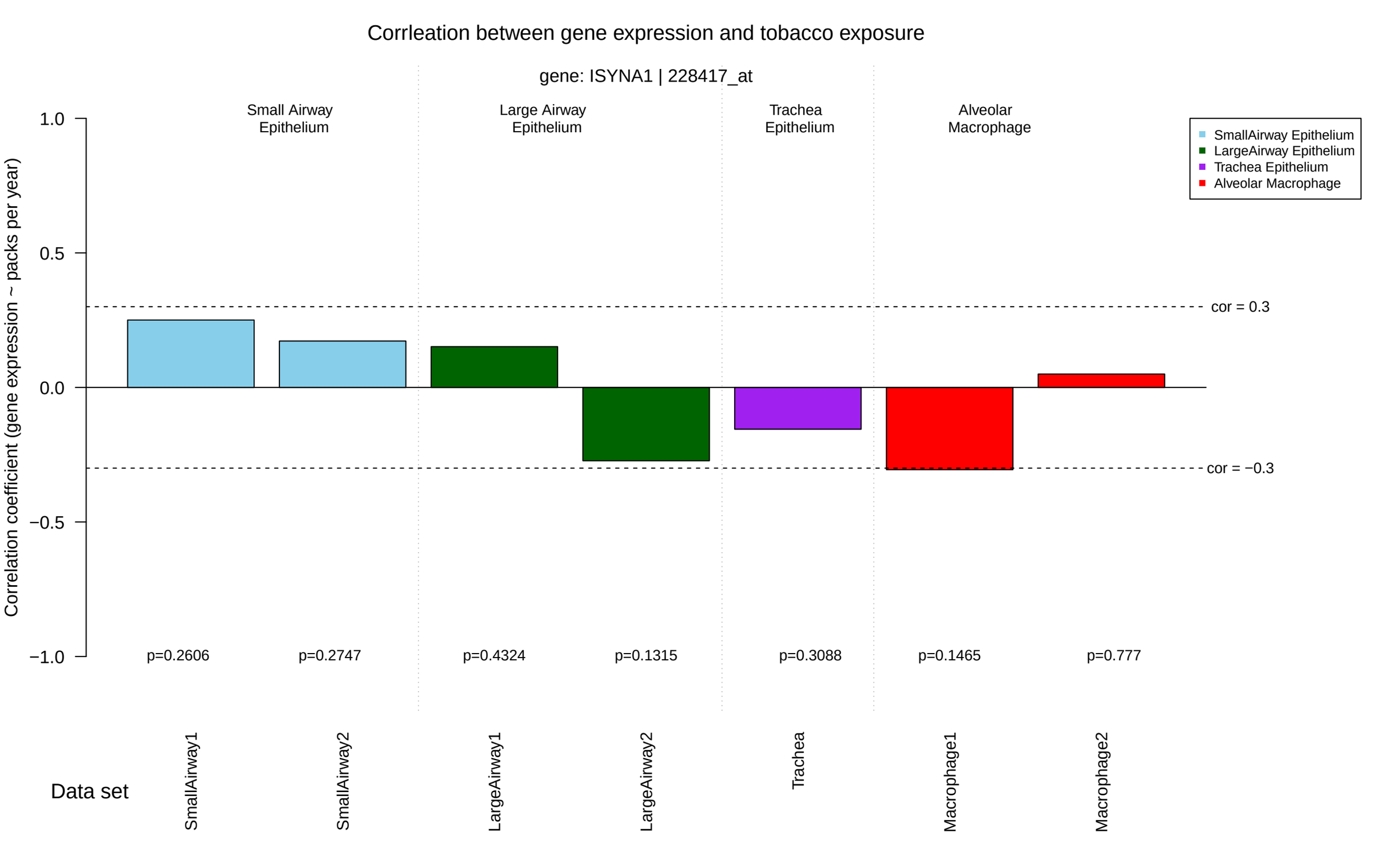

set: LargeAirway1 (p=0.3152)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7512)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9029)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2483)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4873)
ISYNA1 | 228417_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISYNA1 | 228417_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2606
p=0.2747
p=0.4324
p=0.1315
p=0.3088
p=0.1465
p=0.777
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2